

# Problem Set 3

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## Question 1

### Instructions

We are interested in how governments' management of public resources impacts economic prosperity. Our data come from Alvarez, Cheibub, Limongi, and Przeworski (1996) and is labelled `gdpChange.csv` on GitHub. The dataset covers 135 countries observed between 1950 or the year of independence or the first year for which data on economic growth are available ("entry year"), and 1990 or the last year for which data on economic growth are available ("exit year"). The unit of analysis is a particular country during a particular year, for a total  $> 3,500$  observations.

- Response variable:
  - `GDPWdiff`: Difference in GDP between year  $t$  and  $t-1$ . Possible categories include: "positive", "negative", or "no change"
- Explanatory variables:
  - `REG`: 1=Democracy; 0=Non-Democracy
  - `OIL`: 1=if the average ratio of fuel exports to total exports in 1984-86 exceeded 50%; 0= otherwise

Please answer the following questions:

1. Construct and interpret an unordered multinomial logit with `GDPWdiff` as the output and "no change" as the reference category, including the estimated cutoff points and coefficients.
2. Construct and interpret an ordered multinomial logit with `GDPWdiff` as the outcome variable, including the estimated cutoff points and coefficients.

## Answers

As a first step, the dataset was inspected. The outcome variable `GDPWdiff` is not categorical as required for the task, but instead measures continuous changes in GDP per capita. Therefore, it was transformed into a categorical variable with three outcomes (positive, negative, and no change). While this transformation entails a loss of information, it is assumed to be appropriate for the purposes of the exercise.

It is worth noting that there are very few observations in the “no change” category (only 16), making it a relatively small group in the data. This may affect the stability of the estimated model.

The transformation was implemented using the following code:

```
1 gdp_data <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/  
  StatsII_2026/main/datasets/gdpChange.csv", stringsAsFactors = F)  
2  
3 gdp_data$GDPWdiff_cat <- ifelse(gdp_data$GDPWdiff > 0, "positive",  
4                               ifelse(gdp_data$GDPWdiff < 0, "negative",  
5                                     ", "no change"))  
6  
7 gdp_data$GDPWdiff_cat <- factor(  
8   gdp_data$GDPWdiff_cat,  
9   levels = c("negative", "no change", "positive"))  
10 table(gdp_data$GDPWdiff_cat)
```

### Subquestion 1

The first task requires the estimation of an unordered multinomial logit model. However, the categories of `GDPWdiff` (negative, no change, positive) are inherently ordered. Ignoring this ordering may reduce efficiency, but it is assumed that this specification is chosen for pedagogical purposes.

To address this subquestion, an unordered multinomial logit model was estimated with “no change” as the reference category. Table 1 below presents the results of this model.

The intercepts (3.805 for negative growth and 4.534 for positive growth) serve as category-specific cutoff points, representing the baseline log-odds of each outcome relative to no change when all predictors equal zero.

Democracy is positively associated with both negative and positive growth outcomes relative to no change. Exponentiating the coefficients, democracies have approximately four times higher odds of negative growth and nearly six times higher odds of

Table 1: Unordered Multinomial Logit Model (GDPWdiff)

|                      | Negative vs No Change | Positive vs No Change |
|----------------------|-----------------------|-----------------------|
| Intercept            | 3.805***<br>(0.271)   | 4.534***<br>(0.269)   |
| Democracy (REG)      | 1.379*<br>(0.769)     | 1.769**<br>(0.767)    |
| Oil Dependence (OIL) | 4.784<br>(6.885)      | 4.576<br>(6.885)      |
| Observations         | 3721                  |                       |
| Residual Deviance    | 4678.77               |                       |
| AIC                  | 4690.77               |                       |

Standard errors in parentheses

\* p&lt;0.10, \*\* p&lt;0.05, \*\*\* p&lt;0.01

positive growth compared to non-democracies. The effect of democracy on positive growth is statistically significant ( $p = 0.021$ ), while its effect on negative growth is only marginally significant ( $p = 0.073$ ). This suggests that democratic regimes are more strongly and reliably associated with positive economic outcomes, although they are also linked to a higher likelihood of economic fluctuations more generally.

Oil dependence is also associated with higher odds of both positive and negative growth relative to no change. However, despite the large estimated coefficients, the associated standard errors are extremely large, and the effects are not statistically significant ( $p = 0.487$  and  $p = 0.506$ , respectively). This indicates substantial uncertainty in the estimates, and no reliable conclusions can be drawn about the impact of oil dependence on economic growth outcomes.

Overall, the results suggest that democracy is associated with a higher likelihood of economic change (particularly positive growth) relative to stagnation, while the effects of oil dependence remain highly uncertain due to imprecise estimation.

All of this was done using the following code:

```

1 ## make no change ref category
2 gdp_data$GDPWdiff_cat <- relevel(gdp_data$GDPWdiff_cat, ref = "no
  change")
3
4 ## run unordered model
5 library(nnet)
6 mnl_model <- multinom(GDPWdiff_cat ~ REG + OIL, data = gdp_data)
7 summary(mnl_model)

```

```

8
9 z_vals <- summary(mnl_model)$coefficients / summary(mnl_model)$standard
  .errors
10 p_vals <- 2 * (1 - pnorm(abs(z_vals)))
11 p_vals

```

## Subquestion 2

To answer Subquestion 2, an ordered multinomial logit model was estimated to account for the natural ordering of the outcome variable `GDPWdiff` (negative < no change < positive). This model assumes an underlying latent variable representing economic performance, with observed outcomes determined by estimated cutoff points. Table 2 below presents the results of the model.

Table 2: Ordered Logit Model (GDPWdiff)

|                      | Coefficient          |
|----------------------|----------------------|
| Democracy (REG)      | 0.399***<br>(0.075)  |
| Oil Dependence (OIL) | -0.199*<br>(0.116)   |
|                      | Cutoff points        |
| Negative   No Change | -0.731***<br>(0.048) |
| No Change   Positive | -0.711***<br>(0.048) |
| Observations         | 3721                 |
| Residual Deviance    | 4687.689             |
| AIC                  | 4695.689             |

Standard errors in parentheses

\* p<0.10, \*\* p<0.05, \*\*\* p<0.01

The estimated cutoff points are  $-0.731$  (between negative and no change) and  $-0.711$  (between no change and positive). These thresholds partition the latent variable into three regions corresponding to the observed outcomes. Both cutoff points are highly statistically significant ( $p < 0.001$ ), indicating that the categories are well separated on the latent scale.

The coefficient on democracy (REG) is positive and statistically significant ( $p < 0.001$ ). This indicates that democratic regimes are associated with higher values of

the latent economic performance variable. Substantively, this implies that democracies are more likely to experience positive growth and less likely to experience negative growth compared to non-democracies. In terms of odds ratios,  $\exp(0.399) \approx 1.49$ , meaning that democracy increases the odds of being in a higher growth category by approximately 49%.

The coefficient on oil dependence (OIL) is negative and not statistically significant at conventional levels ( $p \approx 0.086$ ). This suggests that oil-dependent countries may be less likely to experience higher growth outcomes, but the evidence is weak and not statistically reliable.

Unlike the unordered multinomial logit model, the ordered logit imposes a single set of coefficients across outcome categories (the proportional odds assumption). This results in a more parsimonious and efficient model, provided that the assumption holds. Overall, the results indicate that democracy is robustly associated with improved economic outcomes, while the effect of oil dependence remains uncertain.

All of this was done using the following code:

```
1 ## ensure correct ordering
2 gdp_data$GDPWdiff_cat <- factor(
3   gdp_data$GDPWdiff_cat,
4   levels = c("negative", "no change", "positive"),
5   ordered = TRUE)
6
7 ## run ordered model
8 library(MASS)
9 ordered_model <- polr(GDPWdiff_cat ~ REG + OIL, data = gdp_data, method
   = "logistic")
10 summary(ordered_model)
11
12 ctable <- coef(summary(ordered_model))
13 p_values <- 2 * (1 - pnorm(abs(ctable[, "t value"])))
14 cbind(ctable, "p value" = p_values)
```

## Question 2

### Instructions

Consider the data set `MexicoMuniData.csv`, which includes municipal-level information from Mexico. The outcome of interest is the number of times the winning PAN presidential candidate in 2006 (`PAN.visits.06`) visited a district leading up to the 2009 federal elections, which is a count. Our main predictor of interest is whether the district was highly contested, or whether it was not (the PAN or their opponents have electoral security) in the previous federal elections during 2000 (`competitive.district`), which is binary (1=close/swing district, 0="safe seat"). We also include `marginality.06` (a measure of poverty) and `PAN.governor.06` (a dummy for whether the state has a PAN-affiliated governor) as additional control variables.

- (a) Run a Poisson regression because the outcome is a count variable. Is there evidence that PAN presidential candidates visit swing districts more? Provide a test statistic and p-value.
- (b) Interpret the `marginality.06` and `PAN.governor.06` coefficients.
- (c) Provide the estimated mean number of visits from the winning PAN presidential candidate for a hypothetical district that was competitive (`competitive.district=1`), had an average poverty level (`marginality.06 = 0`), and a PAN governor (`PAN.governor.06=1`).

### Answers

#### Subquestion a

To address Question 2, part (a), a Poisson regression model was estimated to account for the count nature of the dependent variable. The results of this model are presented in Table 3 below.

The Poisson regression results show that the coefficient for competitive districts is not statistically significant ( $z = -0.477$ ,  $p = 0.634$ ). Therefore, there is no evidence that PAN presidential candidates visit swing districts more frequently than non-competitive districts.

Table 3: Poisson Regression: PAN Campaign Visits

|                      | <i>Dependent variable:</i>  |
|----------------------|-----------------------------|
|                      | Number of PAN Visits (2006) |
| Competitive District | −0.081<br>(0.171)           |
| Marginality (2006)   | −2.080***<br>(0.117)        |
| PAN Governor (2006)  | −0.312*<br>(0.167)          |
| Constant             | −3.810***<br>(0.222)        |
| Observations         | 2,407                       |
| Log Likelihood       | −645.606                    |
| Akaike Inf. Crit.    | 1,299.213                   |
| <i>Note:</i>         | *p<0.1; **p<0.05; ***p<0.01 |

**Subquestion b**

The coefficient on *marginality.06* is negative and highly statistically significant ( $p < 0.001$ ), indicating that municipalities with higher levels of poverty receive fewer campaign visits from PAN presidential candidates. Substantively, this suggests that candidates are less likely to allocate campaign resources to more marginalised areas. Given the log-link function in the Poisson model, the magnitude of the effect is substantial: a one-unit increase in marginality is associated with a large decrease in the expected number of visits, holding other variables constant.

The coefficient on *PAN.governor.06* is also negative, with a  $p$ -value of 0.062, indicating marginal statistical significance at the 10% level. This suggests that municipalities located in states governed by the PAN may receive fewer campaign visits compared to those in states governed by other parties. One possible interpretation is that candidates prioritise areas where the party lacks incumbency advantages, although the evidence for this effect is relatively weak.

**Subquestion c**

To obtain the estimated mean number of visits, we plug the relevant values into the Poisson model:

$$\begin{aligned}\hat{\mu} = & \exp(\beta_0 \text{ (Intercept)}) \\ & + \beta_1 \text{ (competitive.district)} \cdot 1 \\ & + \beta_2 \text{ (marginality.06)} \cdot 0 \\ & + \beta_3 \text{ (PAN.governor.06)} \cdot 1\end{aligned}$$

Using the estimated coefficients:

$$\hat{\mu} = \exp(-3.81023 - 0.08135 - 0.31158) = \exp(-4.20316) \approx 0.015$$

Thus, the estimated mean number of visits for a competitive district with average marginality and a PAN governor is approximately 0.015.